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CMAI, AICTE & RGPV
द्वारा पुरस्कृत

INSTITUTE OF TECHNOLOGY AND MANAGEMENT

DEPT. OF ELECTRONICS & COMMUNICATION

MINOR PROJECT-02 :- TRAIN TRACK FAULT DETECTION ROBOT

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Contents

Problems We Face

Introduction

A Train Track Detection Robot is an automated system designed to monitor the condition of railway tracks. Its main purpose is to detect faults such as cracks, gaps, or obstacles on the tracks.

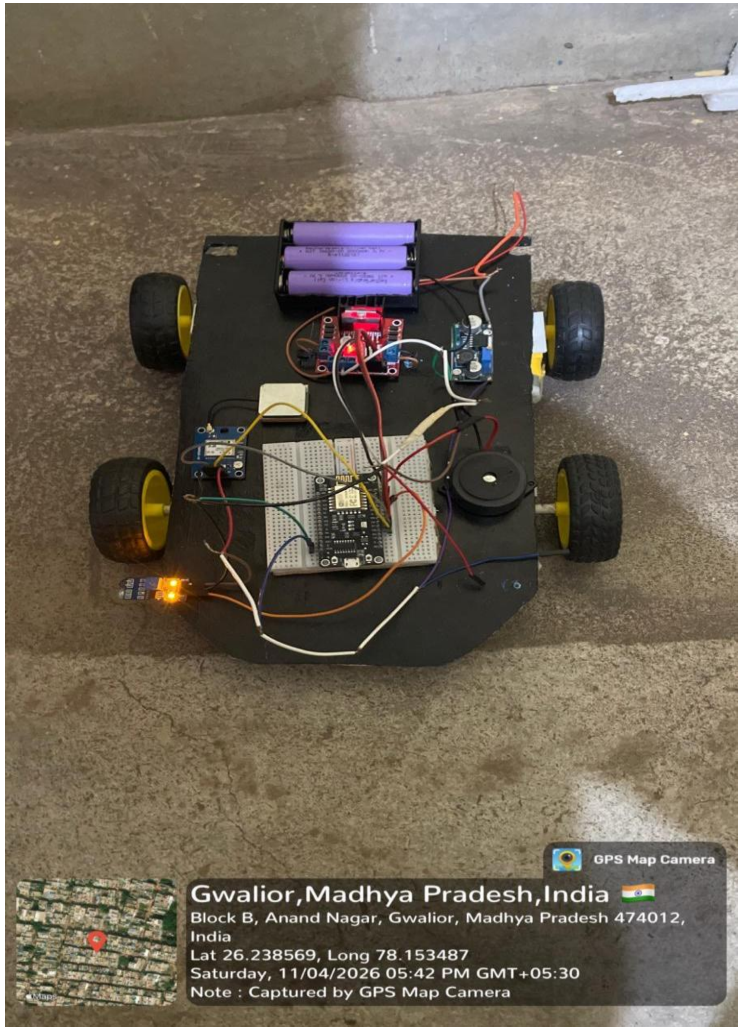
In traditional systems, track inspection is done manually, which is time-consuming and sometimes unsafe. This robot helps to solve that problem by continuously checking the track using sensors.

The robot uses components like IR sensors, ultrasonic sensors to detect any abnormal condition. If a fault is found, it sends an alert to the authorities.

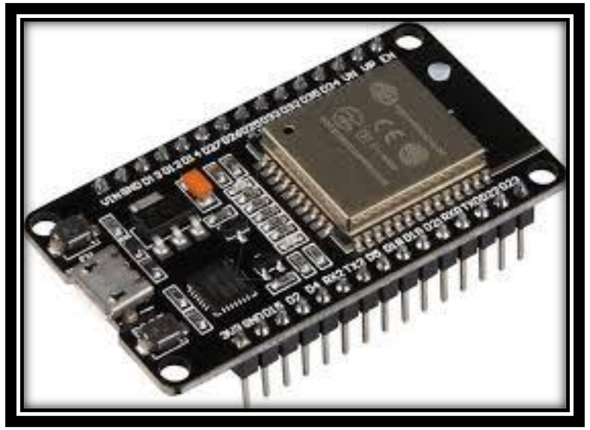
This system improves railway safety, reduces human effort, and helps prevent accidents.

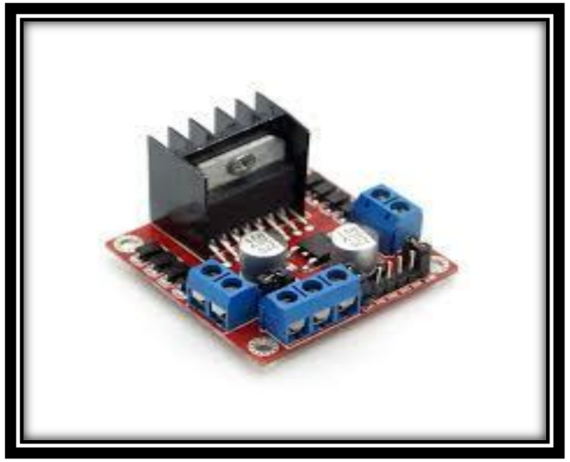
Component Required

- NODE MCU MICROCONTROLLER
- MOTOR DRIVER (L298N)
- DC MOTOR
- IR SENSOR
- GPS MODULE
- BUZZER
- BUCK CONVERTOR
- LI- ION BATTERIES



Modules Used

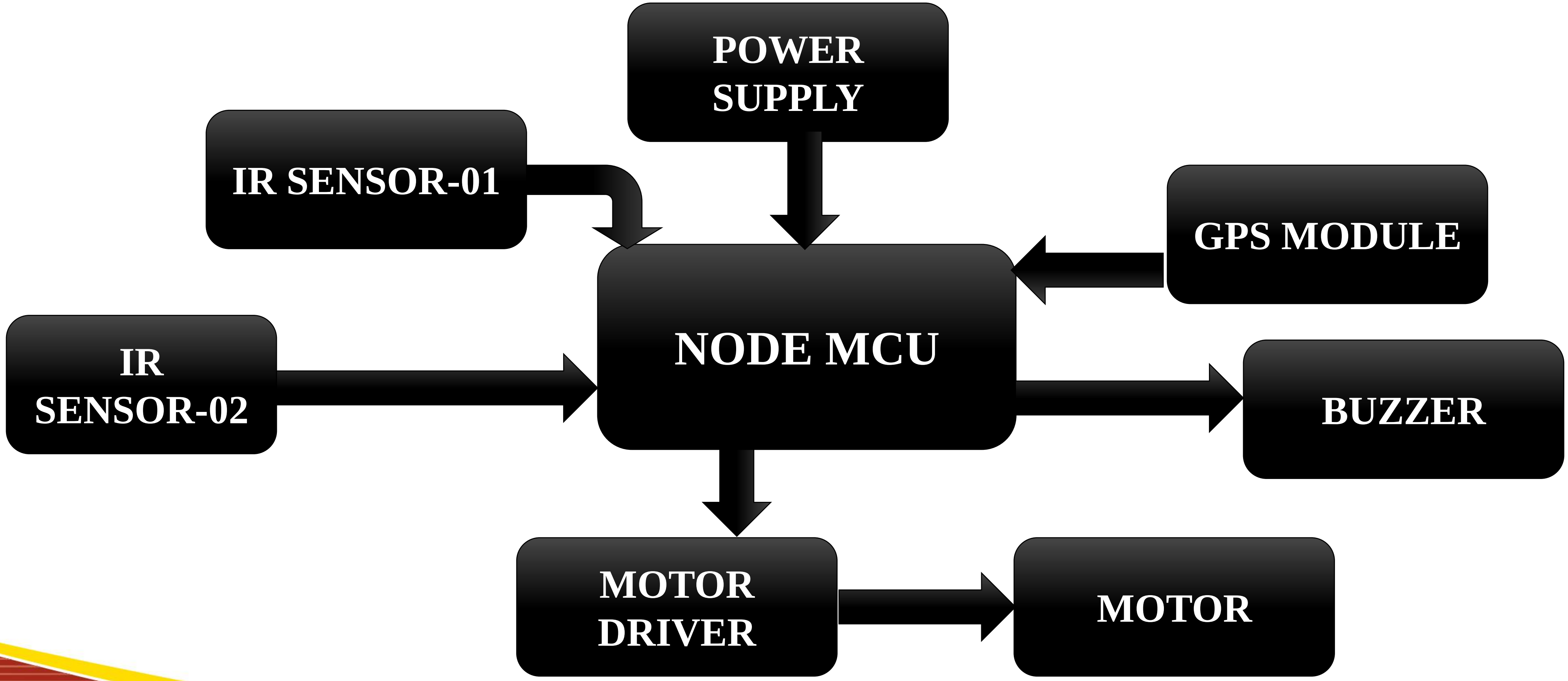




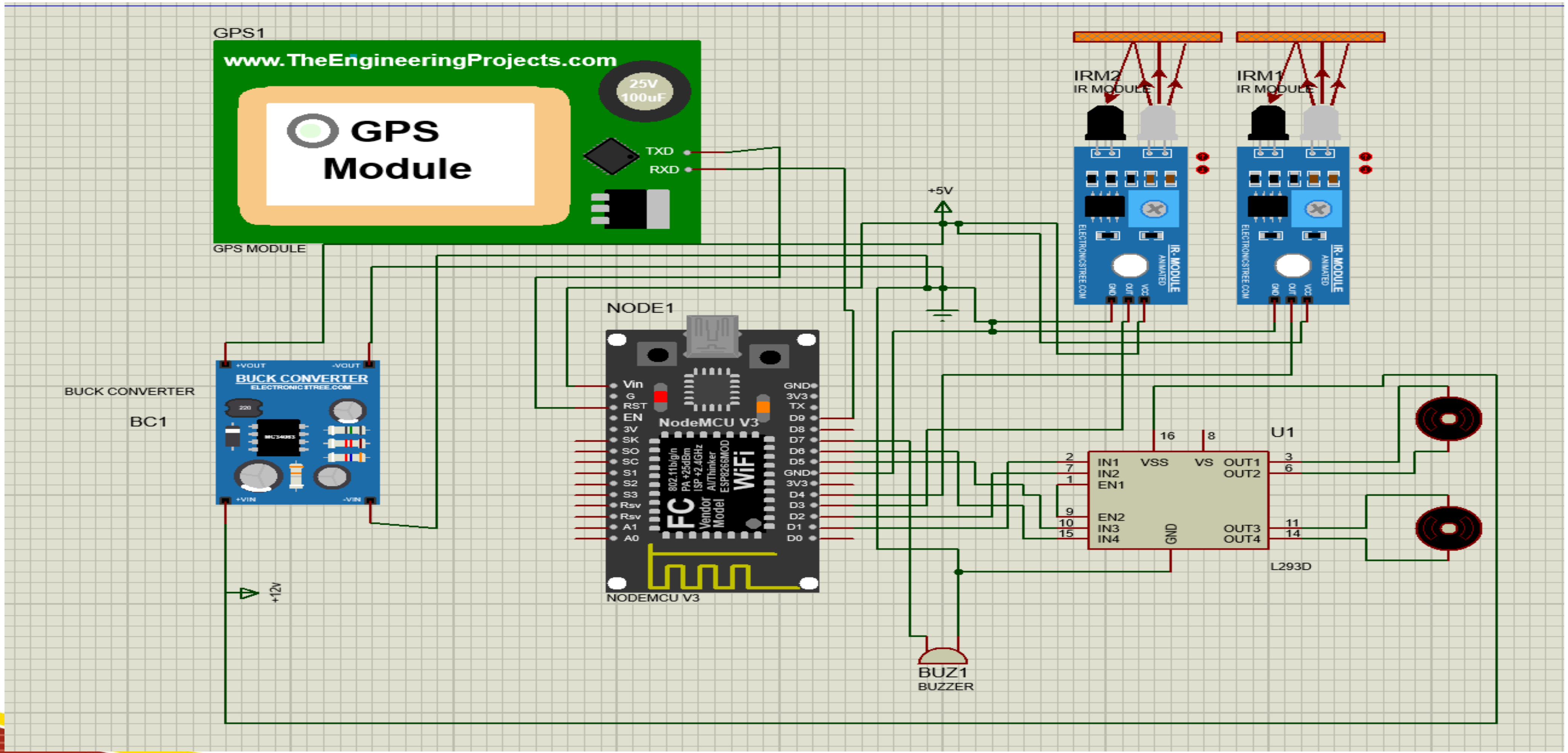
Key Features

- ❑Alert System
- ❑Fault Detection
- ❑Reduced Human Effort
- ❑Real-Time Monitoring
- ❑Autonomous Operation
- ❑Wireless Communication
- ❑Fast Inspection

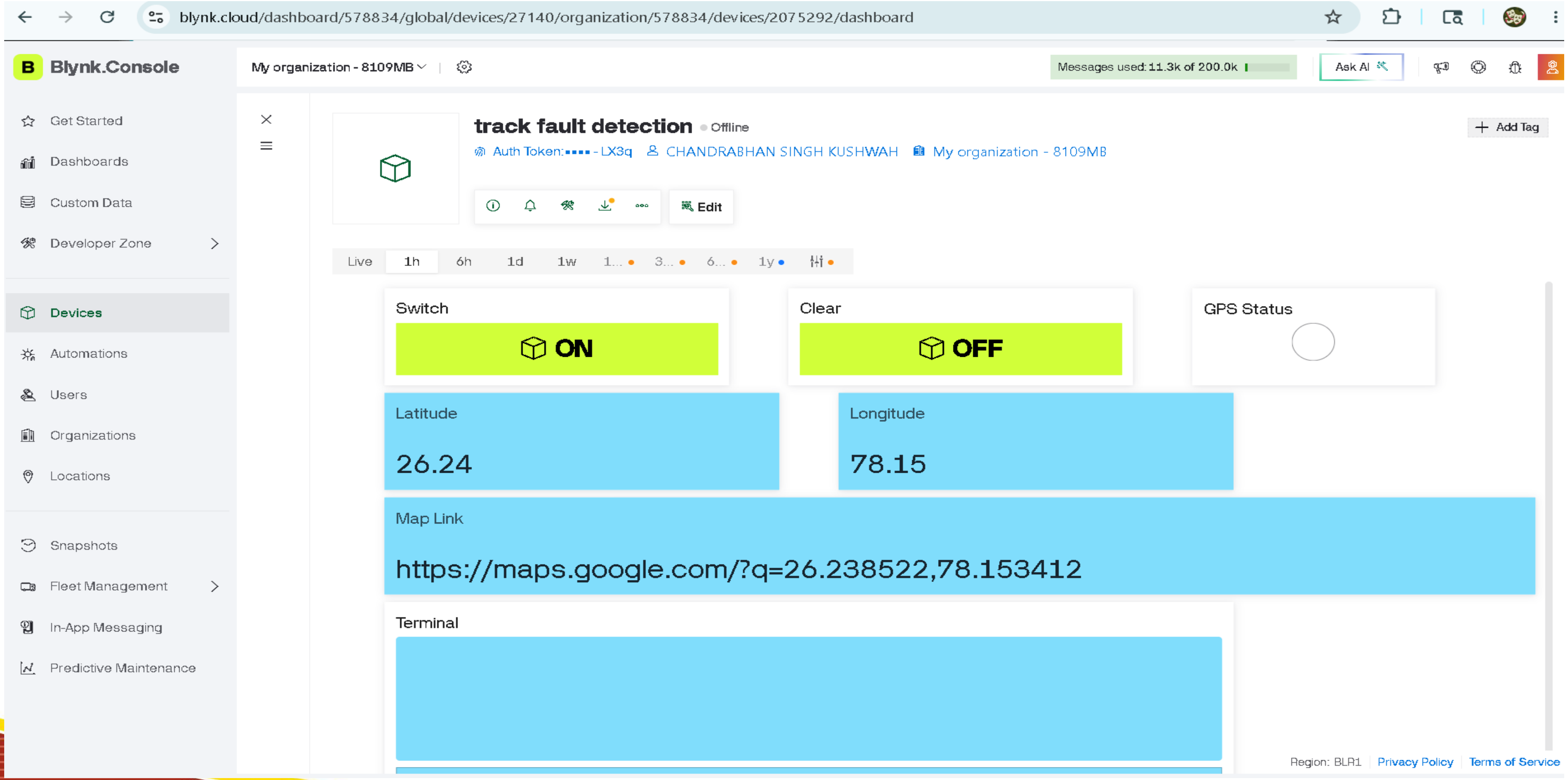
Block Diagram



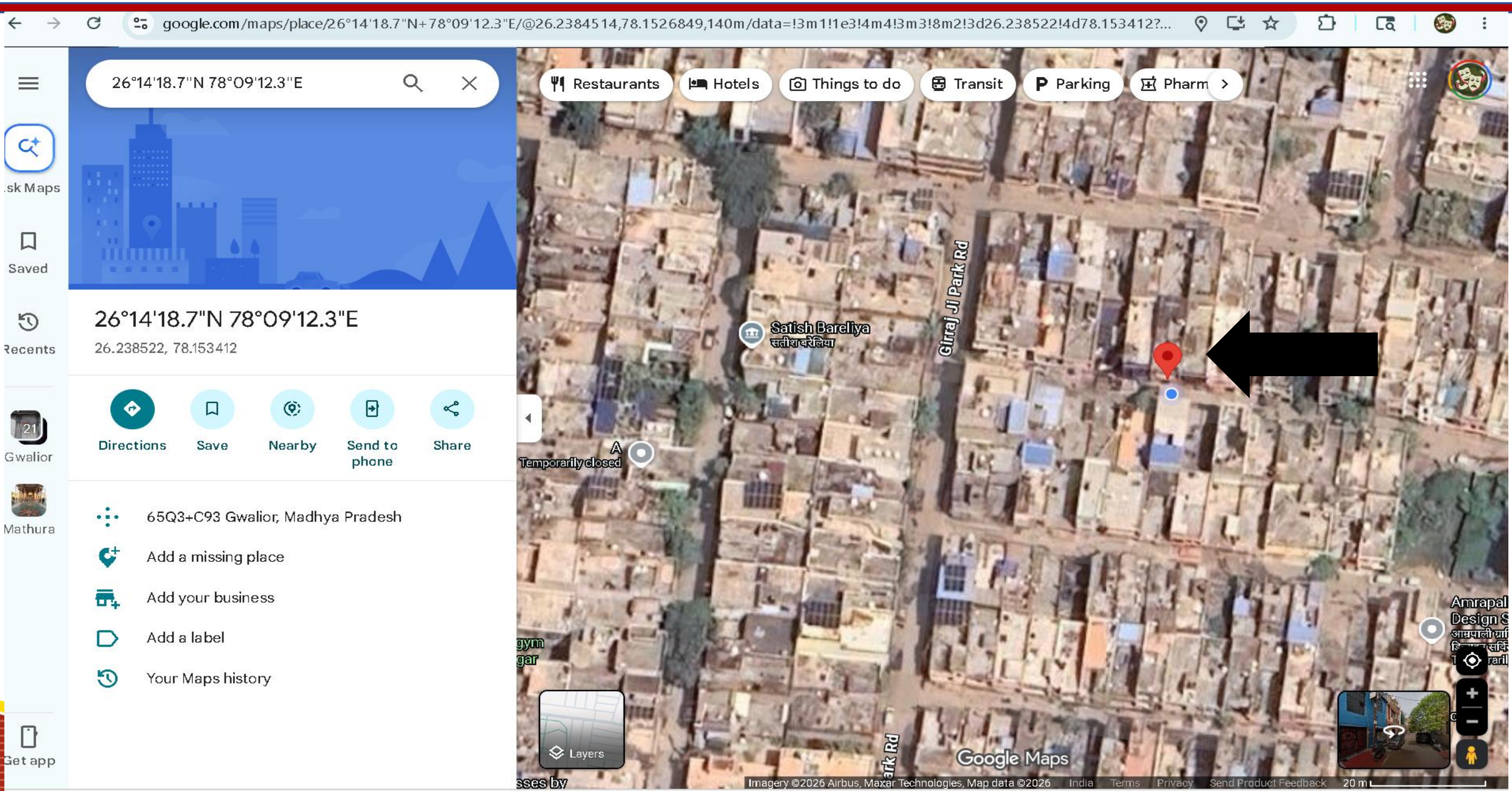
Circuit Diagram



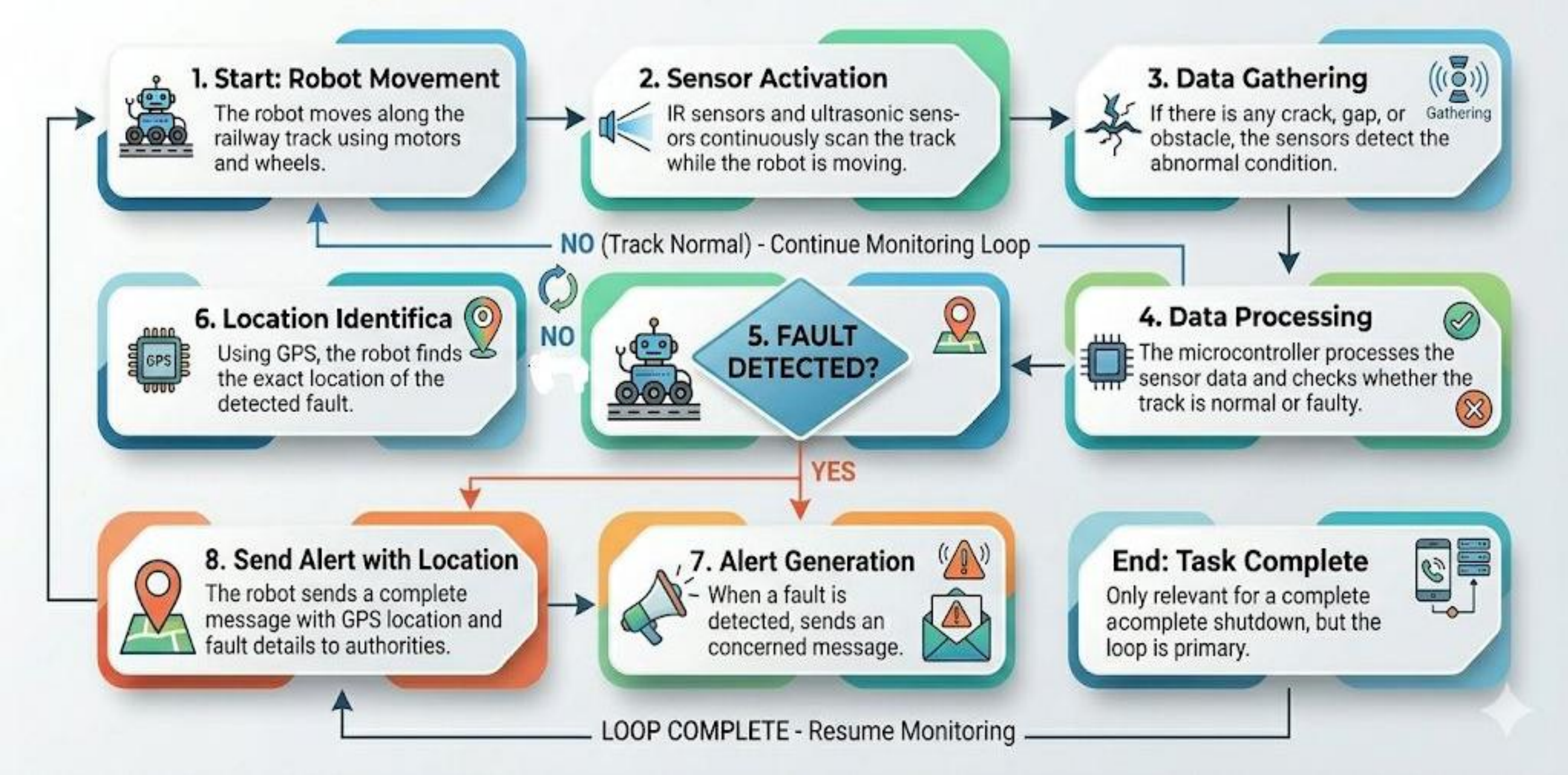
Live GPS and System status Tracking



Location

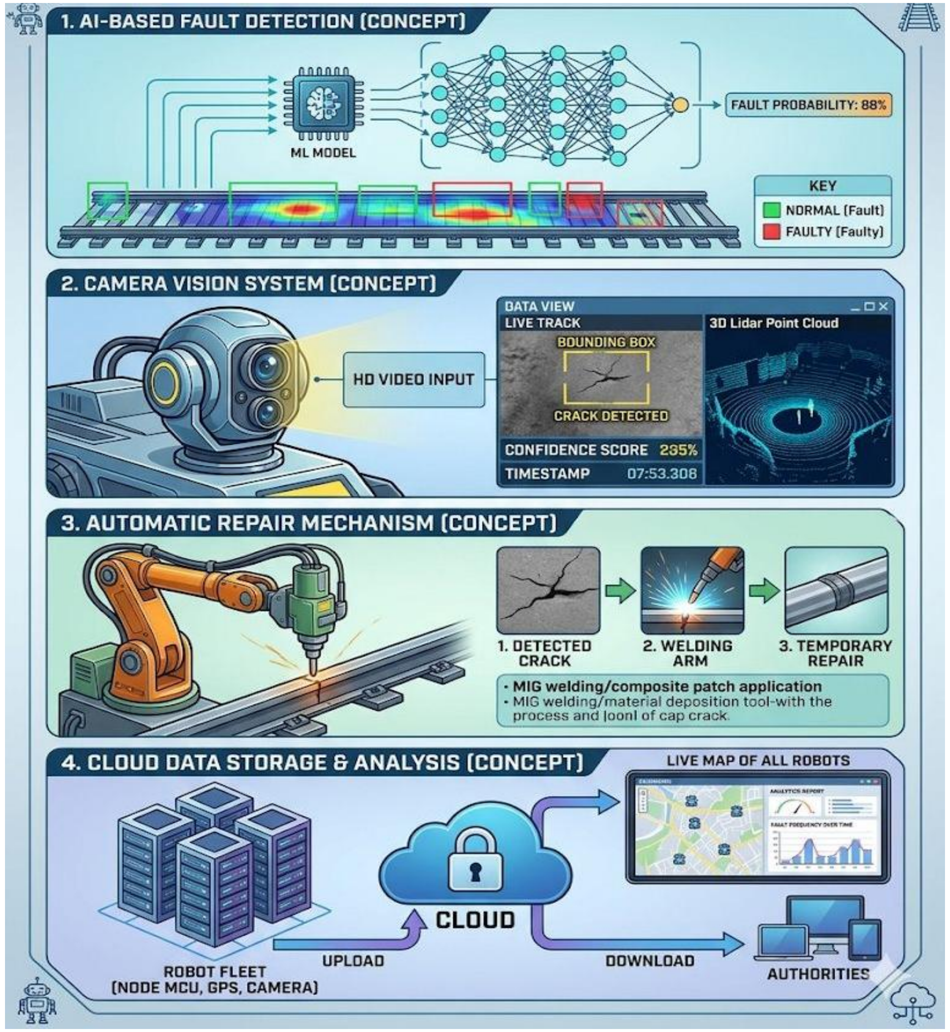

google.com/maps/place/26°14'18.7"N+78°09'12.3"E/@26.2384514,78.1526849,140m/data=!3m1!1e3!4m4!3m3!8m2!3d26.238522!4d78.153412?...
26°14'18.7"N 78°09'12.3"E
Restaurants
Hotels
Things to do
Transit
Parking
Saved
Recents
Gwalior
Mathura
26°14'18.7"N 78°09'12.3"E
26.238522, 78.153412
Directions
Save
Nearby
Send to phone
Share
65Q3+C93 Gwalior, Madhya Pradesh
Add a missing place
Add your business
Add a label
Your Maps history
Satish Bareliya
सतीश बरेलिया
Giriraj Ji Park Rd
Temporarily closed
Layers
Google Maps
Imagery ©2026 Airbus, Maxar Technologies, Map data ©2026
India
Terms
Privacy
Send Product Feedback
20 m

Working


1. Start: Robot Movement
The robot moves along the railway track using motors and wheels.
2. Sensor Activation
IR sensors and ultrasonic sensors continuously scan the track while the robot is moving.
3. Data Gathering
Gathering
If there is any crack, gap, or obstacle, the sensors detect the abnormal condition.
NO (Track Normal) - Continue Monitoring Loop
4. Data Processing
The microcontroller processes the sensor data and checks whether the track is normal or faulty.
5. FAULT DETECTED?
NO
YES
6. Location Identifica
GPS
Using GPS, the robot finds the exact location of the detected fault.
7. Alert Generation
When a fault is detected, sends an concerned message.
8. Send Alert with Location
The robot sends a complete message with GPS location and fault details to authorities.
End: Task Complete
Only relevant for a complete acomplete shutdown, but the loop is primary.
LOOP COMPLETE - Resume Monitoring

Advancements

- AI-Based Fault Detection.
- Camera Vision System.
- Automatic Repair Mechanism.
- Cloud Data Storage & Analysis.
- Solar Charging.



Conclusion

The Train Track Detection Robot is an effective system for improving railway safety by detecting faults like cracks, gaps, and obstacles on the tracks.

It reduces the need for manual inspection and provides faster and more reliable monitoring.

By using technologies like sensors, GPS, and solar power, the system becomes efficient and suitable for real-time applications.

Overall, this project helps in preventing accidents, saving time, and making railway maintenance smarter and safer.


THANK YOU